

CHAPTER 2

SURVEY OF TECHNOLOGIES

2.1 SURVEY OF TECHNOLOGY

2.1.1 FRONTEND

The development of the Self-Drive Car Booking System involves the use of modern web technologies to ensure the platform is secure, responsive, scalable, and user-friendly. The technologies used in both frontend and backend components contribute to the system's functionality and performance.

HTML (HYPERTEXT MARKUP LANGUAGE)

- HTML provides the structural framework for the web application, allowing developers to organize content such as forms, buttons, booking tables, and user dashboards.
- Its tag-based syntax ensures accessibility, semantic structure, and compatibility across all modern browsers.
- HTML is the foundation for integrating styles (CSS) and behavior (TypeScript), offering a clean and maintainable structure.

ADVANTAGES:

- Simple and easy to learn
- Compatible with all browsers
- Provides a semantic and organized content structure
- Integrates easily with CSS and JavaScript

DISADVANTAGES:

- Static by default; cannot handle dynamic behavior alone
- Depends on CSS and JavaScript for styling and interactivity
- Limited functionality for complex user interfaces.

CSS (CASCADING STYLE SHEETS)

- CSS styles the visual components of the application, including layout, colors, fonts, and responsiveness.
- It separates presentation from structure, promoting cleaner code and reusability across multiple pages.
- With media queries, CSS ensures the application is fully responsive and adapts well to both desktop and mobile devices.

ADVANTAGES:

- Enables consistent styling across all pages
- Supports responsive design through media queries
- Reduces redundancy by using external stylesheets
- Easy to update the look and feel of the site

DISADVANTAGES:

- Can become complex to manage in large projects without proper structure
- Browser compatibility issues may arise
- No built-in logic (unlike JavaScript)

TYPESCRIPT

- TypeScript adds interactivity to the platform, handling client-side validation, live content updates, and user interactions.
- As a client-side scripting language, it enhances the user experience by reducing server load and enabling instant feedback.
- TypeScript operates within the browser and allows the development of a seamless, dynamic user experience.

ADVANTAGES:

- Adds type safety to JavaScript, reducing runtime errors
- Improves code readability and maintainability in large projects
- Supports object-oriented programming concepts such as interfaces, enums, and generics.
- Seamlessly integrates with modern JavaScript libraries.
- Compiles to clean, browser-compatible JavaScript

DISADVANTAGES:

- Requires a compilation step, adding complexity to the build process
- Slightly steeper learning curve for beginners unfamiliar with typed language.

2.1.2 BACKEND – SUPABASE

Supabase serves as the backend-as-a-service (BaaS) platform for the Self Drive Car Booking System. It provides a powerful, scalable alternative to traditional backend development by combining a PostgreSQL database with built-in authentication, storage, and real-time subscriptions.

ADVANTAGES:

- Easy Setup & Scalable: Supabase provides an instant backend with database, auth, and APIs, reducing development time.
- Built-in Auth & RLS: Includes secure user authentication and powerful row-level security policies.
- Realtime Features: Allows live syncing of data changes (e.g., new bookings, availability updates).

DISADVANTAGES:

- Limited Customization vs. Traditional Backend: Some constraints when compared to building a full custom backend from scratch.
- Learning Curve for Policies: Row-level security can be tricky to configure for beginners.

2.1.3 DATABASE – POSTGRESQL

The database layer is managed by Supabase, which uses PostgreSQL, a robust and scalable open-source relational database system.

POSTGRESQL

- PostgreSQL handles the storage of users, projects, contributions, reward tiers, and transaction data.
- It supports complex queries, data relationships, and transactions, which are vital for maintaining data integrity in financial systems.
- PostgreSQL's support for JSON, indexing, and triggers allows flexibility in storing structured and semi-structured data efficiently.
- With Supabase's interface, PostgreSQL is managed easily through a web dashboard, SQL editor, and API layer.

ADVANTAGES:

- Fully ACID-compliant, ensuring data integrity and consistency
- Supports complex queries, joins, triggers, and stored procedures
- Allows relational data modeling with strong support for foreign keys and constraints
- Highly extensible with custom functions, data types, and full-text search

DISADVANTAGES:

- Slightly more complex to configure and optimize than simpler databases like SQLite or MySQL
- Requires knowledge of SQL for effective querying and database management
- Can consume more system resources compared to lighter databases